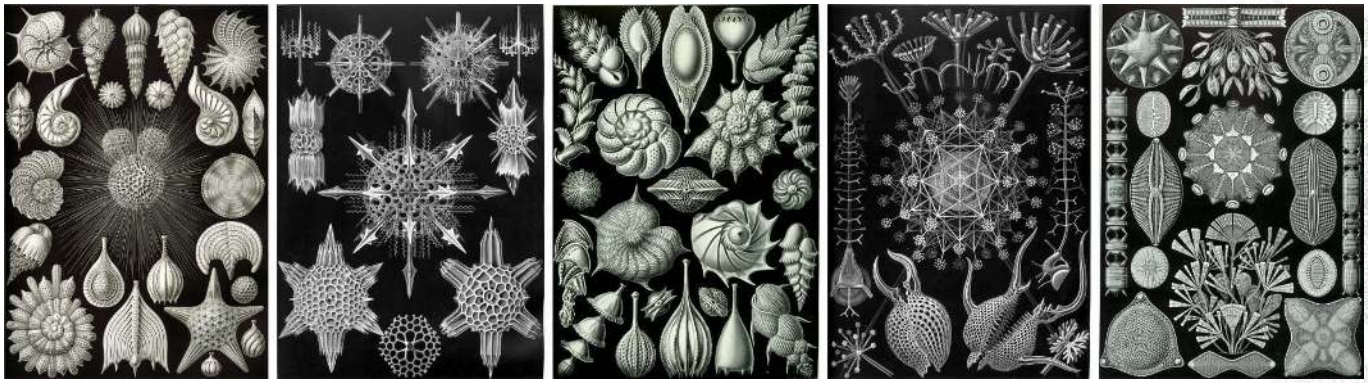
EART27201 Sedimentary Rocks and Fossils



Microfossils

Russell Garwood | russell.garwood@manchester.ac.uk

Contents:

This lecture covers the topics:

-- An introduction to microfossils.

-- Key Siliceous Microfossils.

-- Phosphatic Microfossils.

-- Important Calcareous Microfossils.

-- Organic-Walled Microfossils.

What is a microfossil?

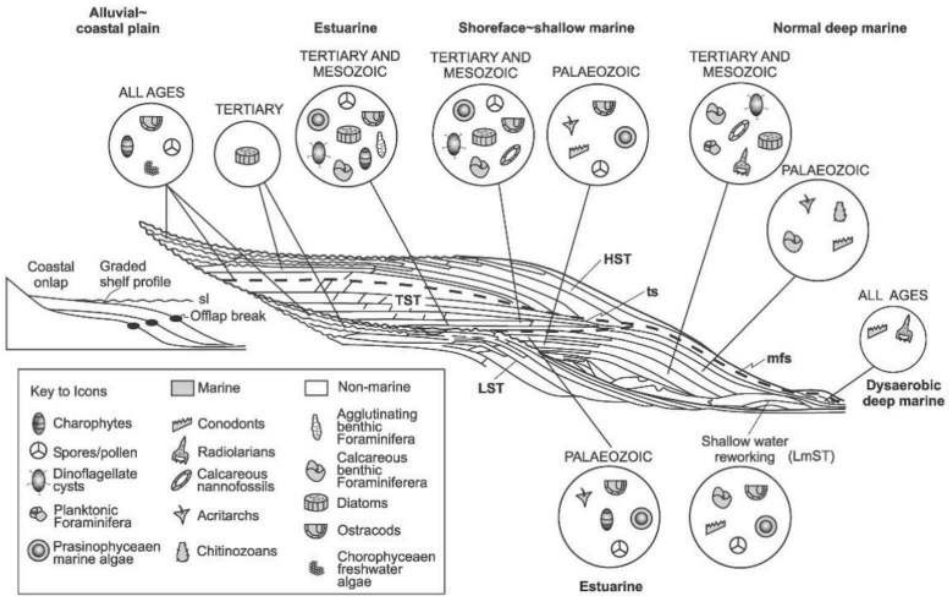


Microfossil: Any fossil that is best studied by means of a microscope. Material may include dissociated fragments of larger organisms, whole organisms of microscopic size, or embryonic forms of larger fossil organisms. Various groups, e.g. acritarchs, Foraminiferida, Ostracoda, and Conodontophora, are studied and used as stratigraphic markers.

Eukaryote: An organism comprising cells that have a distinct nucleus enveloped by a double membrane, and other features including double-membraned mitochondria and 80S ribosomes in the fluid of the cytoplasm (i.e. all protoctists, fungi, plants, and animals).

Source: Allaby, M., 2013. A dictionary of geology and earth sciences. Oxford University Press. Image source: Nautral History Museum, London.

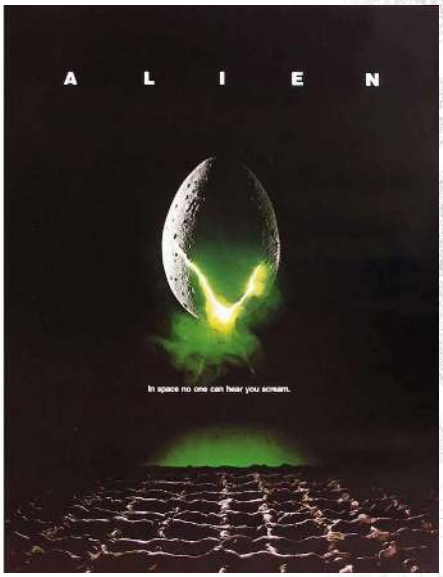
In what environments do these organisms live?



Source: Armstrong, H. and Brasier, M., 2013. Microfossils. John Wiley & Sons.

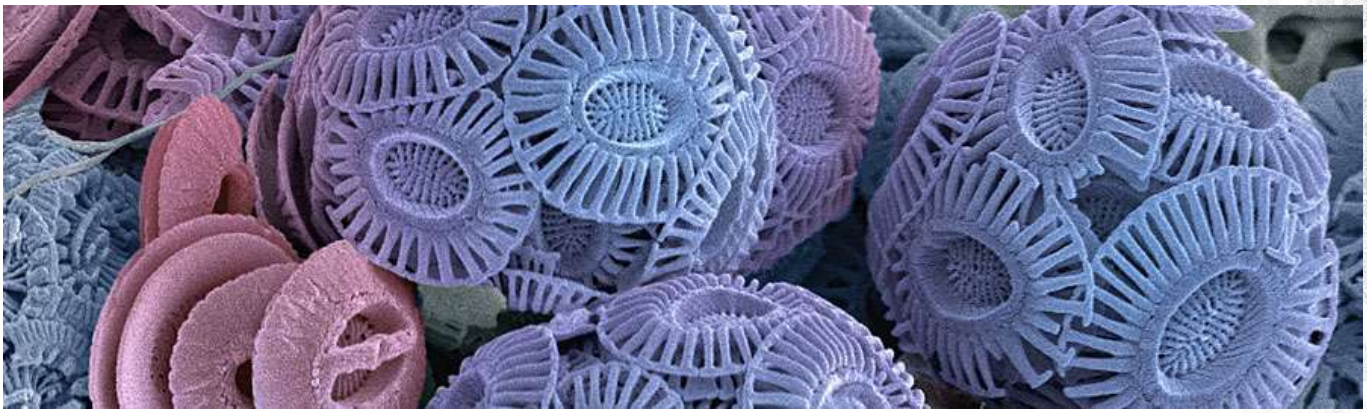
What are the risks?





Scott, R., Gordon, C., Giler, D. and Hill, W., 1979. Alien. United States: Twentieth Century-Fox.

How are microfossils useful?



Coccolithophores



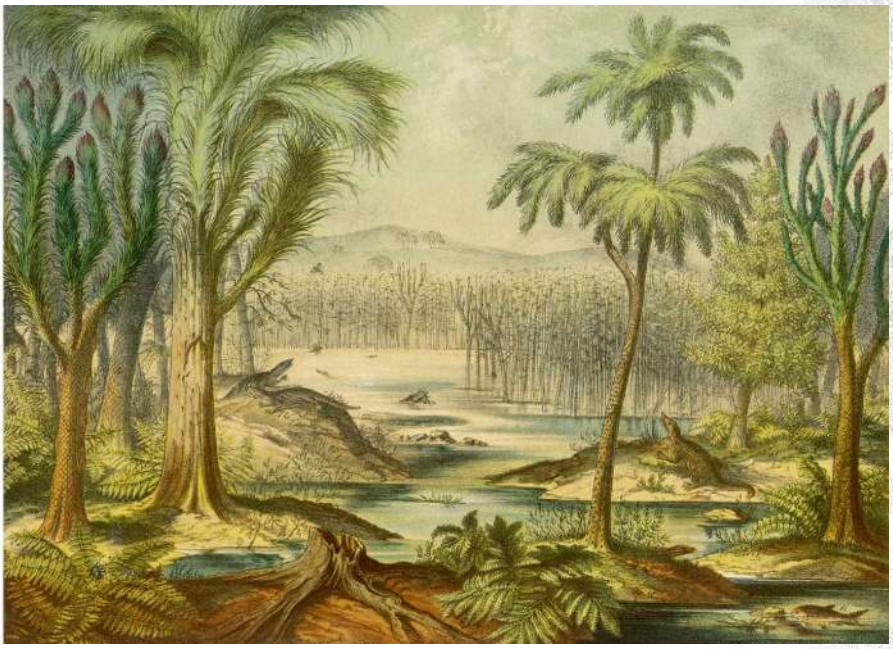
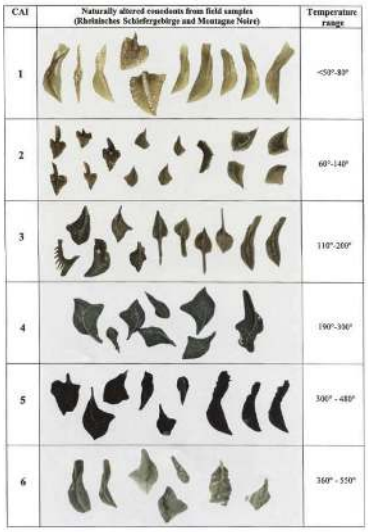
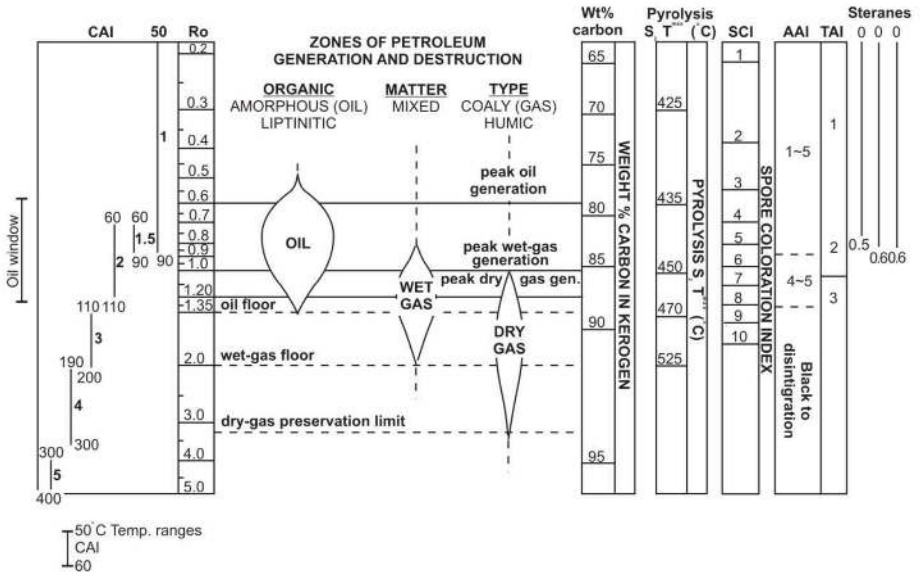




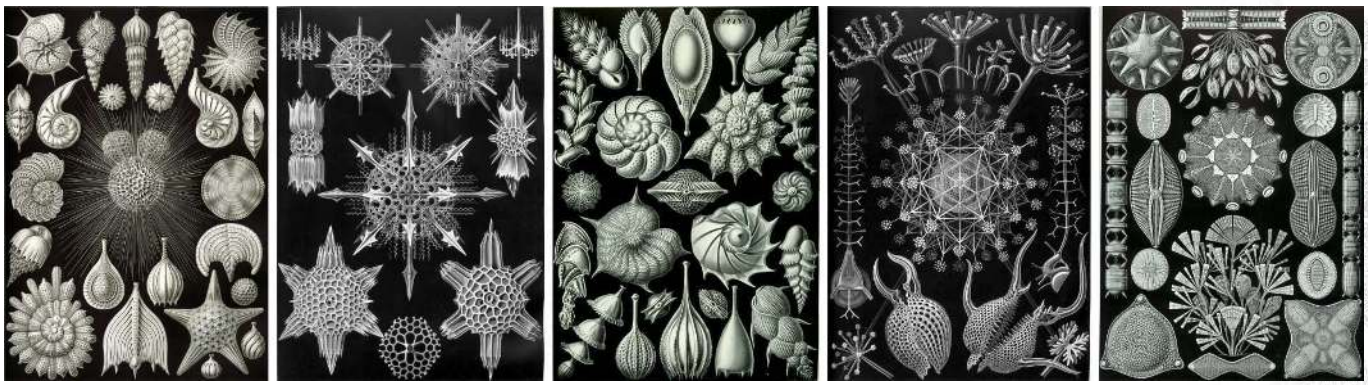
Image: US Geological Survey





Right: Comparision of the main indicators of thermal maturity with the zones of petroleum generation and destruction. AAI, acritarch alteration index; CAI, conodont colour alteration index; SCI, spore coloration index; TAI, thermal alteration index. Source: Armstrong, H. and Brasier, M., 2013. Microfossils. John Wiley & Sons.
Left:CAI, conodont colour alteration index. Source: Königshof, P., 2003. Conodont deformation patterns and textural alteration in Paleozoic conodonts: examples from Germany and France. Senckenbergiana lethaea, 83(1-2), pp.149-156.

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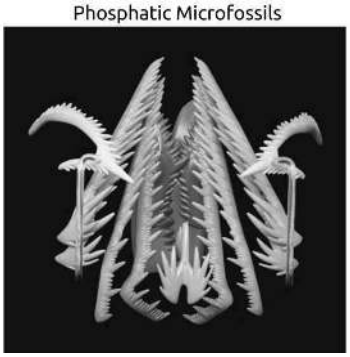
Microfossils

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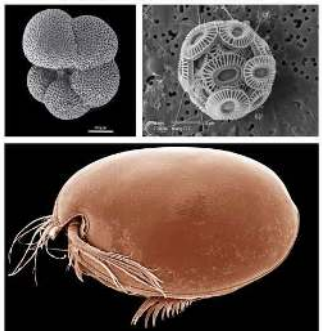
Siliceous Microfossils



Phosphatic Microfossils



Calcareous Microfossils



Organic-Walled Microfossils



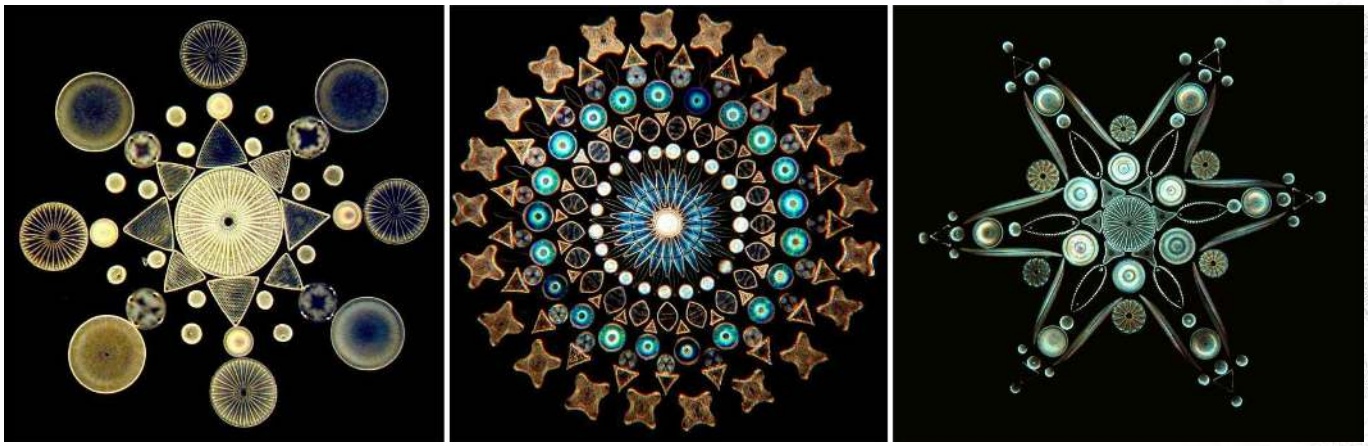
Siliceous Microfossils

-- Diatoms (?Jurassic to Recent)

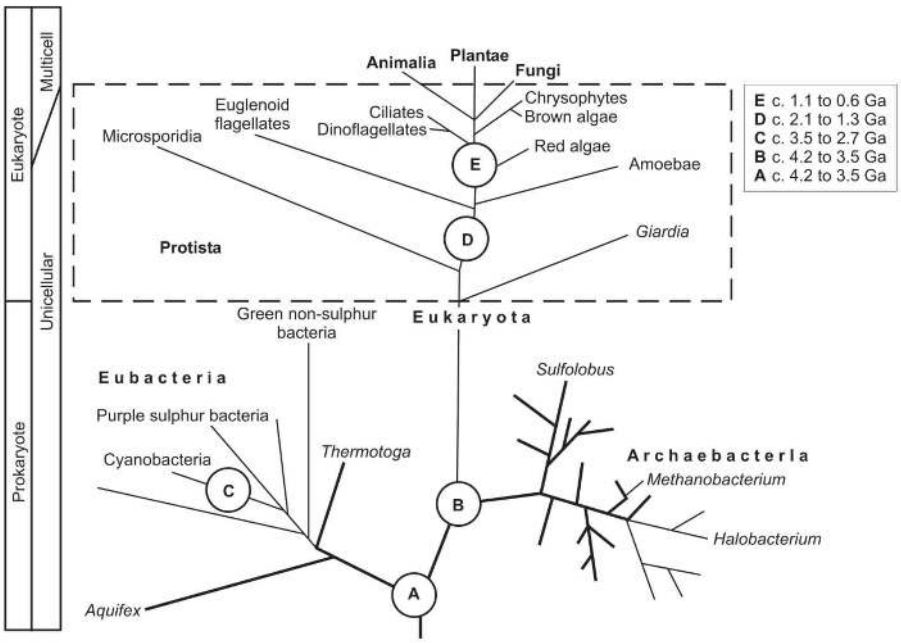
-- Radiolaria (Cambrian to Recent)

-- Silicoflagellates (Late Cretaceous to Recent)

What is a diatom?

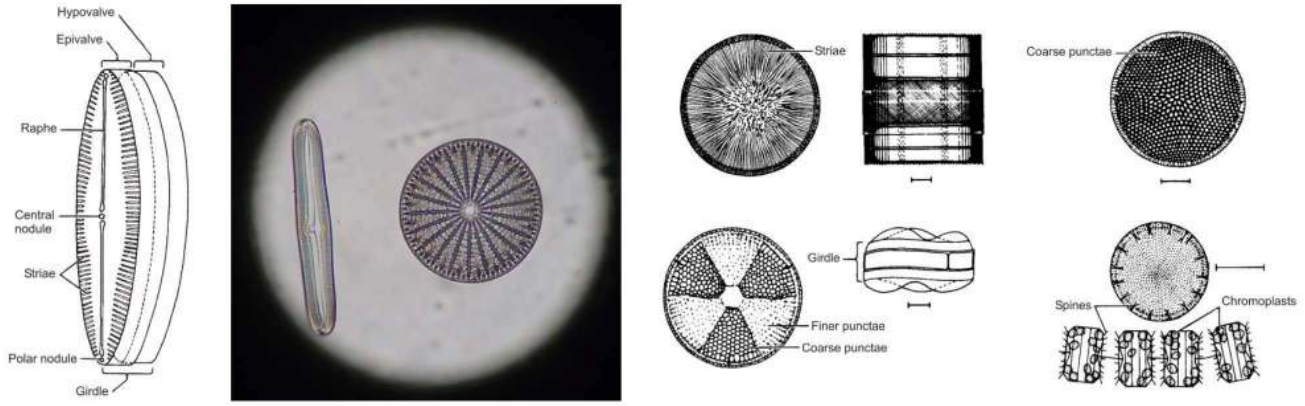


Frustule: The cell wall of a diatom, which is impregnated with silica. The wall consists of two halves that overlap one another.



Chrysophyte: A large phylum (or group, i.e. the chrysophytes) of mostly freshwater photosynthetic protists, including the golden, or golden-brown, algae.

Source: Armstrong, H. and Brasier, M., 2013. Microfossils. John Wiley & Sons.



Scale bars, 10 µm. Source: Armstrong, H. and Brasier, M., 2013. Microfossils. John Wiley & Sons.

Epitheca: In diatoms (Bacillariophyta), the older, outer half of the frustule.

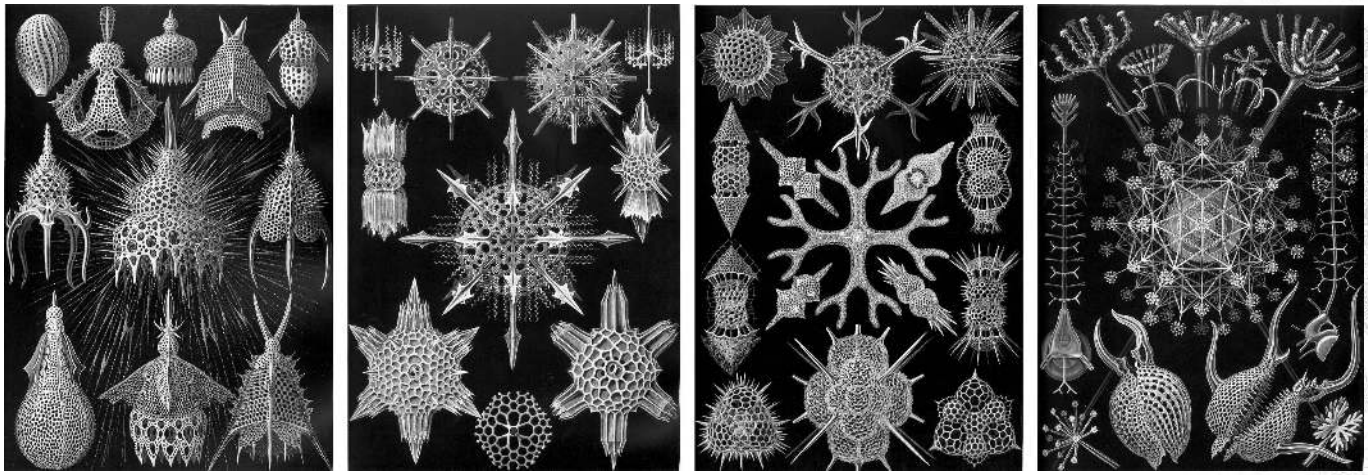
Hypotheca: In diatoms (Bacillariophyta), the younger, inner half of the frustule.

Frustule: The cell wall of a diatom, which is impregnated with silica. The wall consists of two halves that overlap one another.

Centric diatom: The Centrales (centric diatoms) are circular, with radial symmetry, and are predominantly marine.

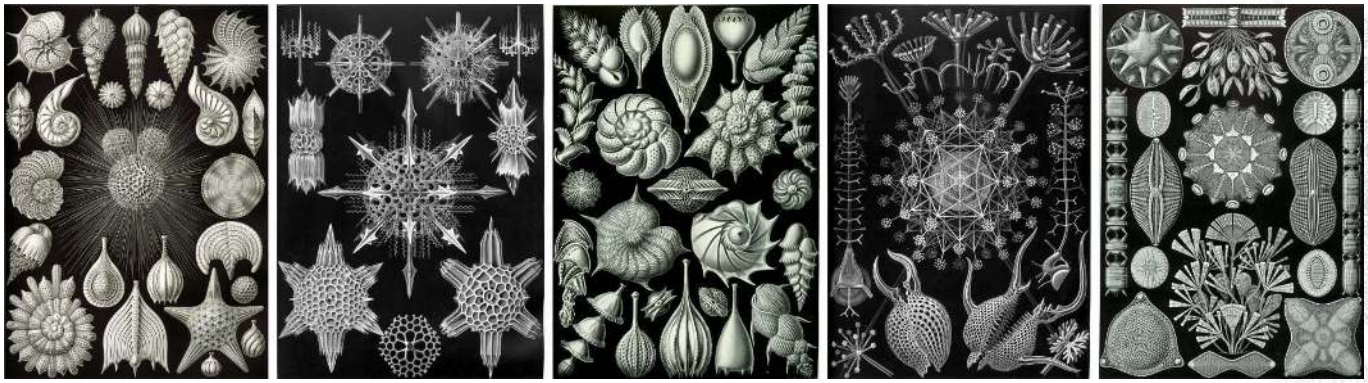
Pinnate diatom: The Pennales (pinnate diatoms) are elliptical, with bilateral symmetry, and dominate freshwater environments.

What is a radiolarian?



Source: Haeckel, E., 1904. Kunstformen der Natur. Die einhundert Tafeln.

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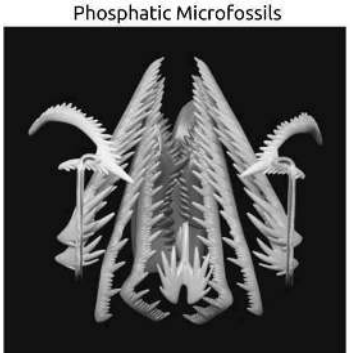
Microfossils

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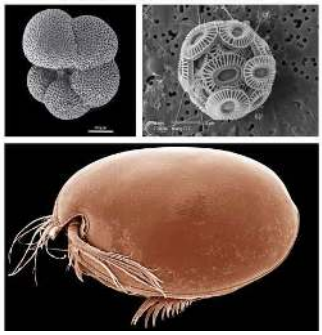
Siliceous Microfossils



Phosphatic Microfossils



Calcareous Microfossils



Organic-Walled Microfossils

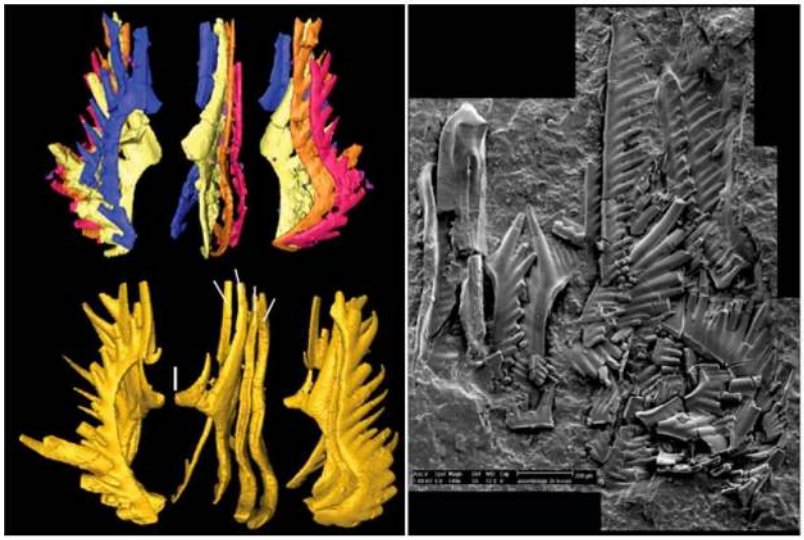


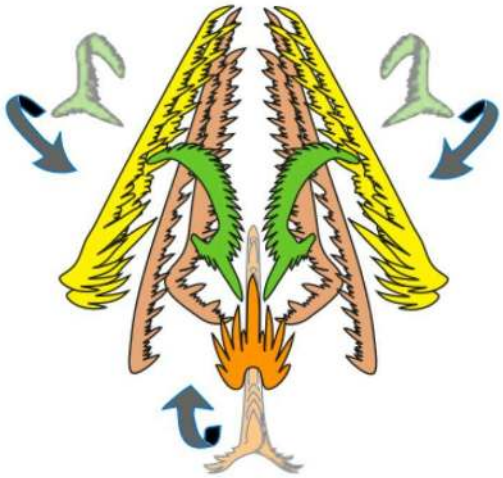
Phosphatic Microfossils

-- Conodonts (Cambrian to Triassic)

-- Fish/shark teeth (Cambrian to Recent)
-- Some scolecodonts (Cambrian to Recent)

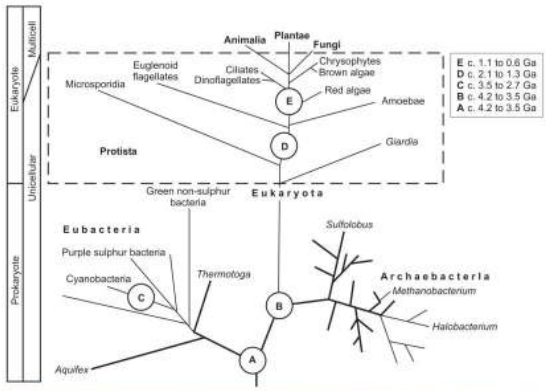
What is a conodont?



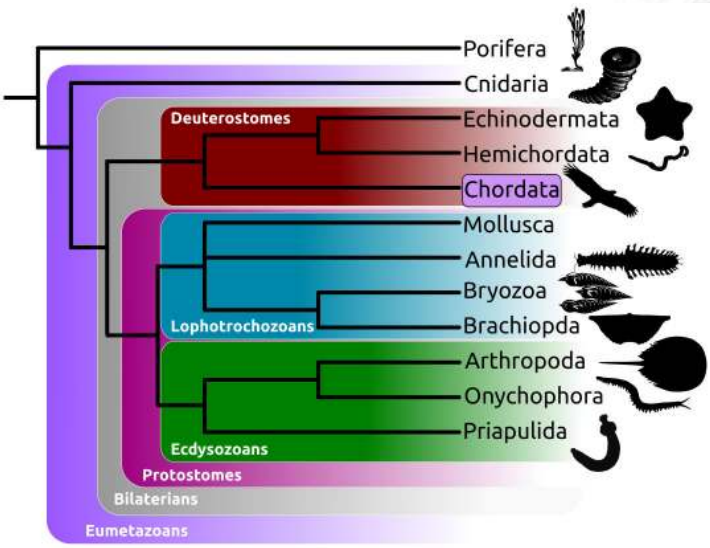


Calcium carbonate fluorapatite: $Ca_5(PO_4,CO_3)_3(F,OH,O)$

Goudemand, N., Orchard, M.J., Urdy, S., Bucher, H. and Tafforeau, P., 2011. Synchrotron-aided reconstruction of the conodont feeding apparatus and implications for the mouth of the first vertebrates. Proceedings of the National Academy of Sciences, 108(21), pp.8720-8724.



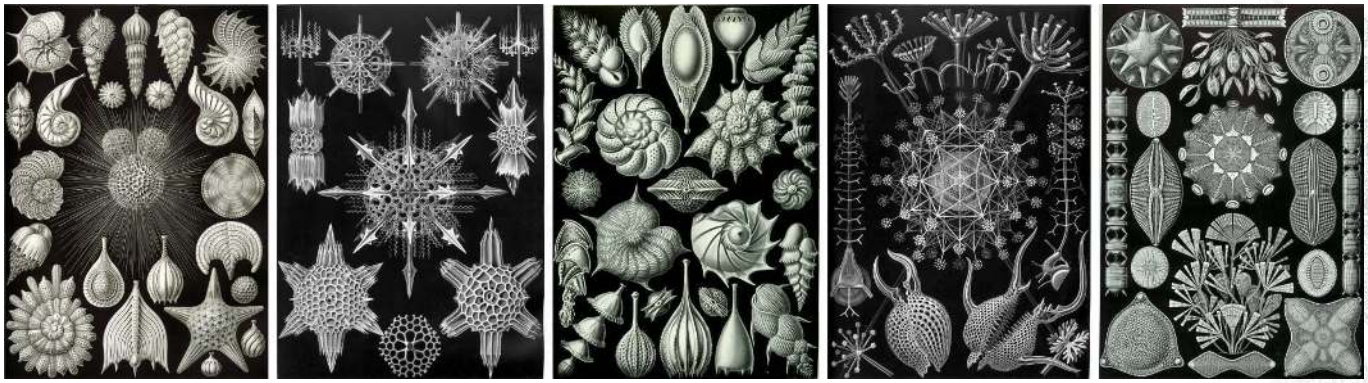




Conodontophorid: Extinct jawless marine animals, known chiefly for their distinctive mineralized mouthparts, which are commonly found as fossils in rock strata dating from the early Cambrian to the end of the Triassic periods (about 525 to 200 mya)

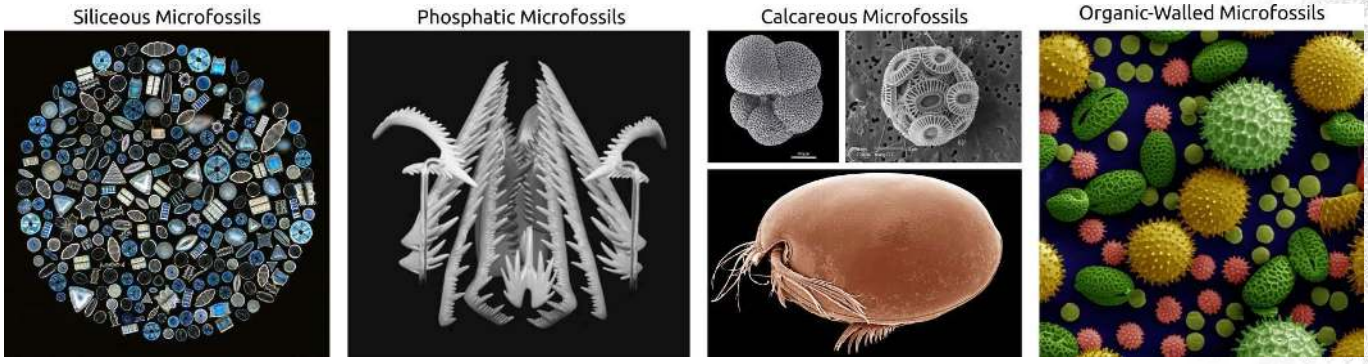
Source: Top left: Armstrong, H. and Brasier, M., 2013. Microfossils. John Wiley & Sons.
Bottom Left: Life restoration of *Promissum pulchrum* by Nobu Tamura.

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Microfossils

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Calcareous Microfossils

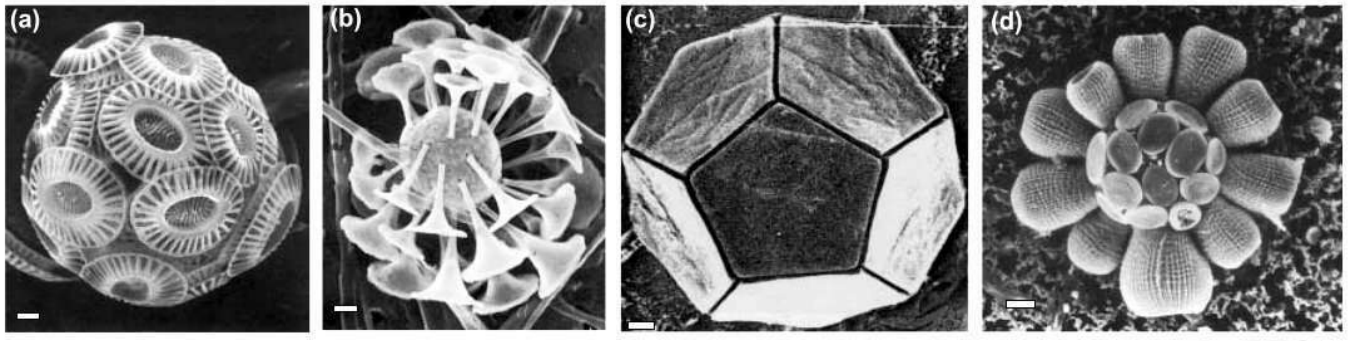
-- Calcareous nannoplankton (Late Triassic to Recent)

-- Foraminifera (Cambrian to Recent)

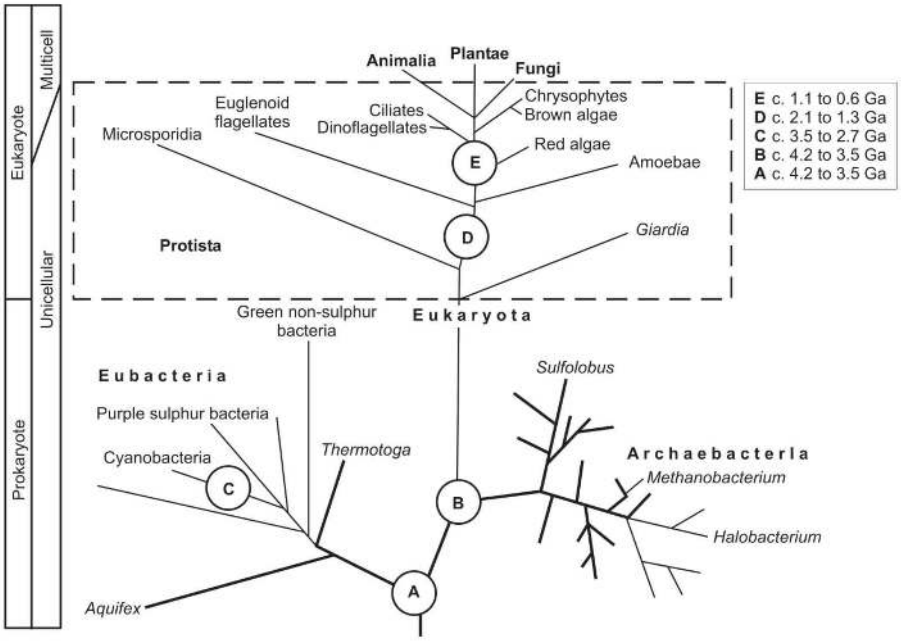
-- Ostracods (Cambrian to Recent)

-- Pteropods (Late Cretaceous to Recent)
-- Calpionellids (Late Jurassic to Early Cretaceous)
-- Calcareous Algae (Precambrian to Recent)

What are calcareous nannoplankton?

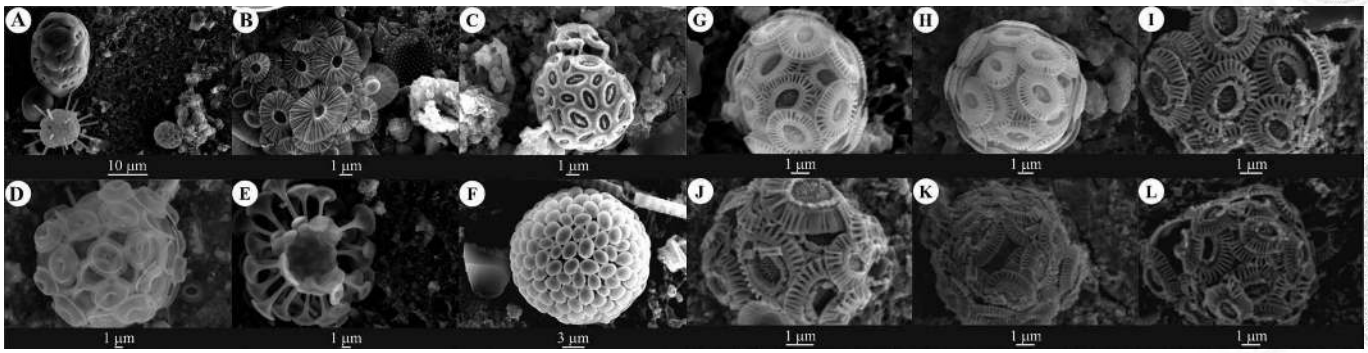


Electron photomicrographs of living coccolithophores. Scale bar = 1 μm. Source: Armstrong, H. and Brasier, M., 2013. Microfossils. John Wiley & Sons.



Phytoplankton: The photosynthetic plankton and primary producers of aquatic ecosystems.

Source: Armstrong, H. and Brasier, M., 2013. Microfossils. John Wiley & Sons.



Coccolith: The microscopic calcareous plates or discs, often oval and commonly intricately patterned and ornamented, that occur as part of the protective covering of a group of the unicellular algae called **coccolithophorids [or coccolithophores]**.

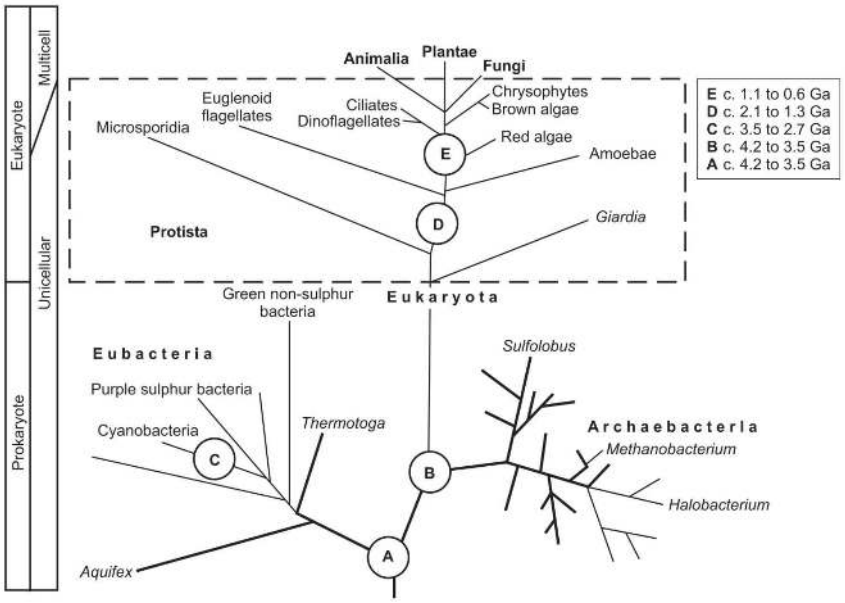
Source: Ziveri, P., Passaro, M., Incarbona, A., Milazzo, M., Rodolfo-Metalpa, R. and Hall-Spencer, J.M., 2014. Decline in coccolithophore diversity and impact on coccolith morphogenesis along a natural CO2 gradient. The Biological Bulletin, 226(3), pp.282-290.

What are Foraminifera?



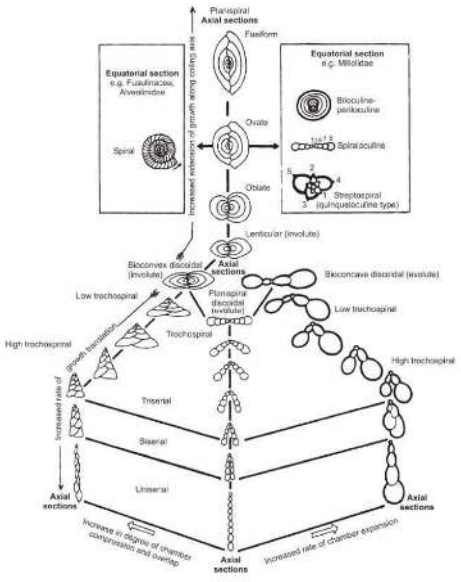
Test = shell.

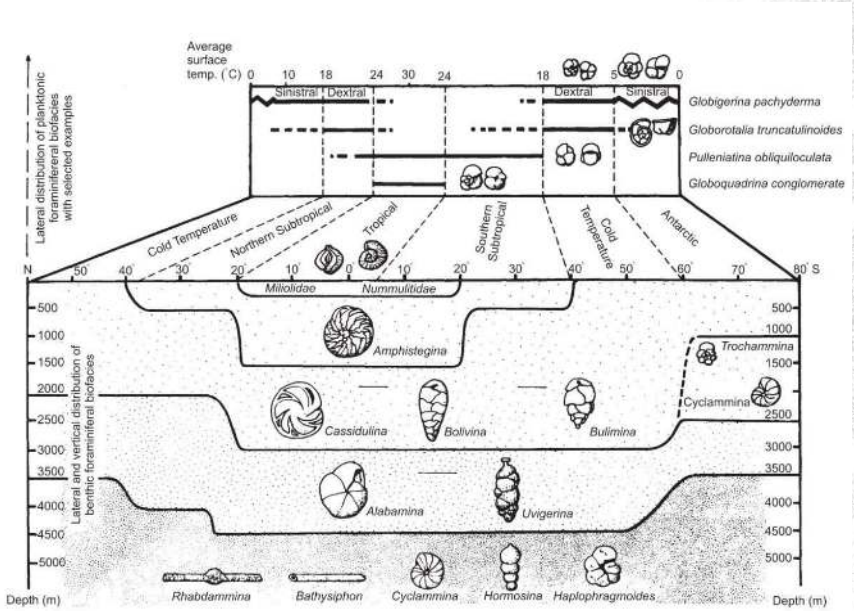
Source: Haeckel, E., 1904. Kunstformen der Natur. Die einhundert Tafeln.



Protozoa: Formerly an animal phylum (Protozoa) of eukaryotic, single-celled microorganisms. Protozoa are now included in the kingdom Protoctista in the five-kingdom classification and in the kingdom Protista (domain Eukarya) in the three-domain classification.
Amoeba: Any single-celled eukaryote that is naked and changes shape due to the irregular extension and retraction of pseudopodia.

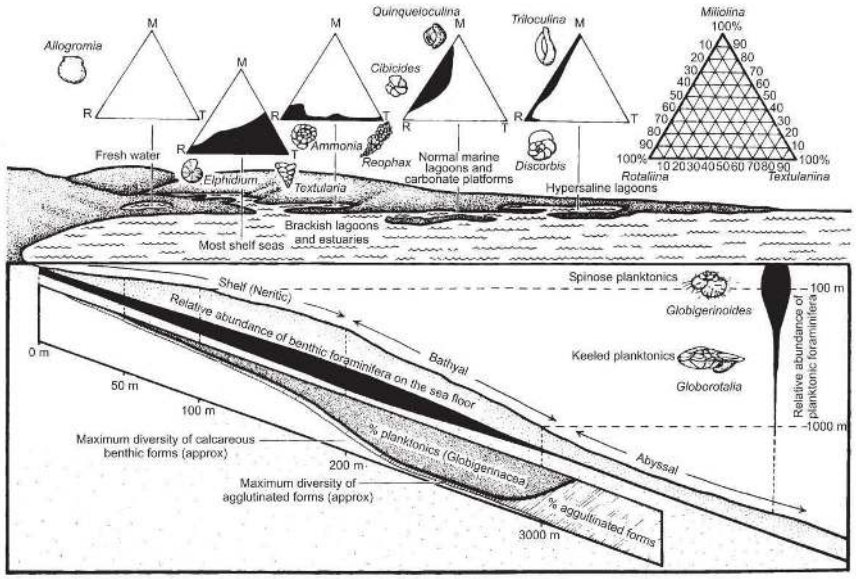
Source: Armstrong, H. and Brasier, M., 2013. Microfossils. John Wiley & Sons.





Source: Armstrong, H. and Brasier, M., 2013. Microfossils. John Wiley & Sons.

How are Foraminifera useful?

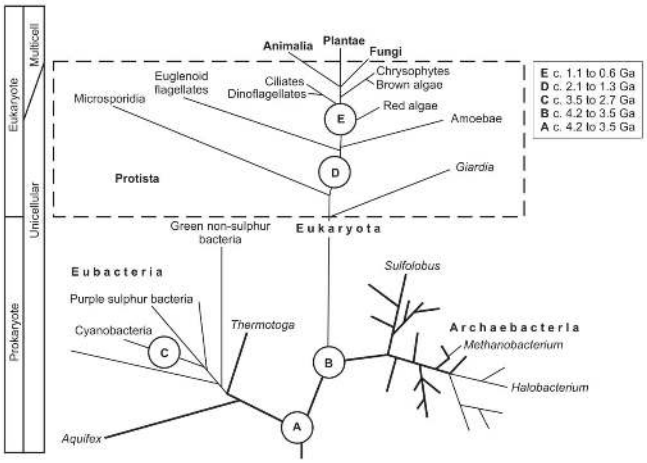


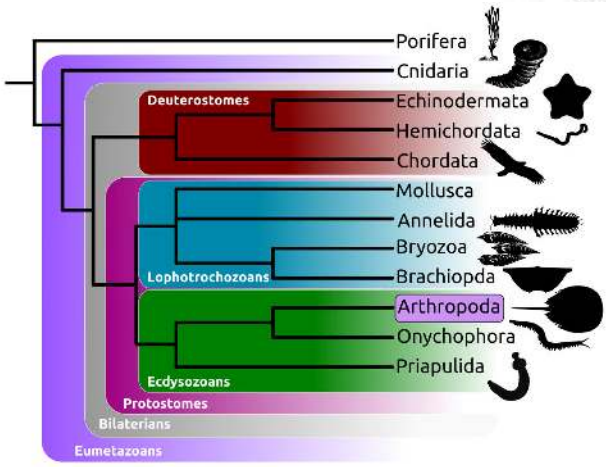
Source: Armstrong, H. and Brasier, M., 2013. Microfossils. John Wiley & Sons.

What is an ostracod?

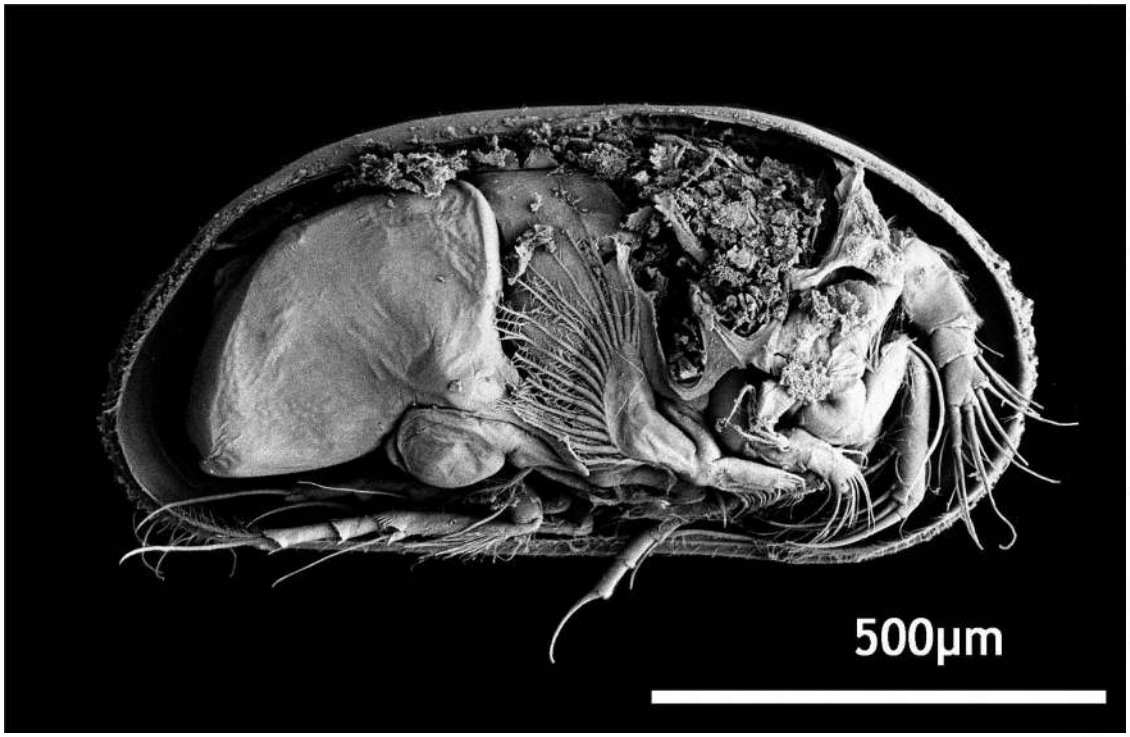


The Carapace of the ostracod *Vargula hilgendorfii*. Image courtesy of Vincent Perrier (distributed via imaggeo.egu.eu).

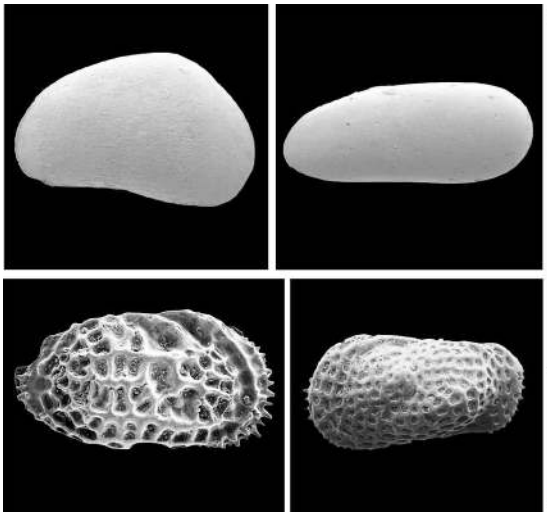


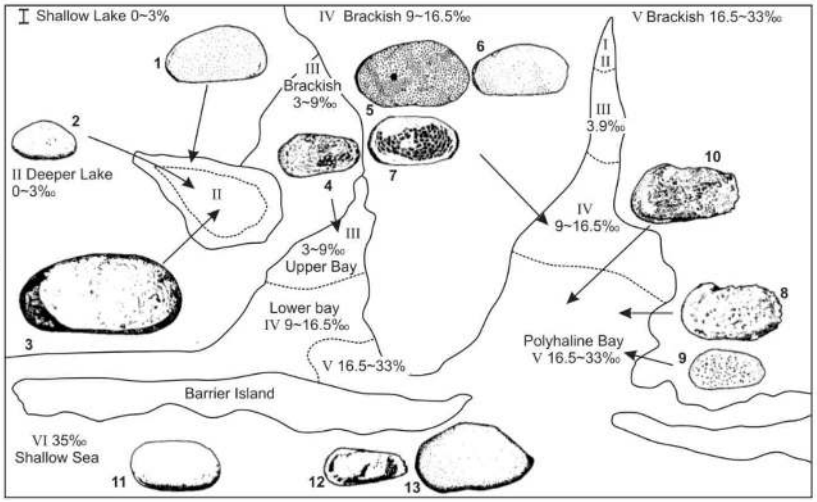


Crustacea: An arthropod group including rabs, lobsters, shrimps, slaters, woodlice, barnacles.



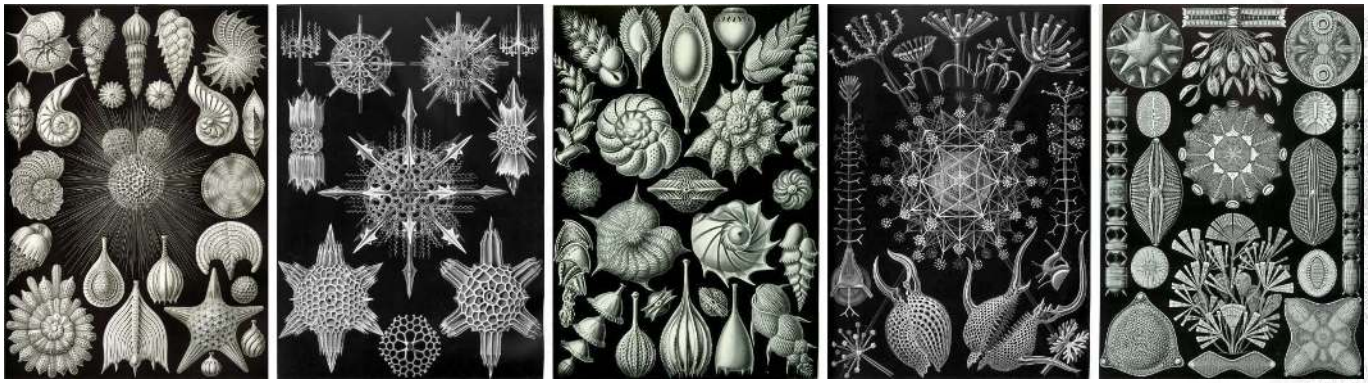
How are ostracods useful?





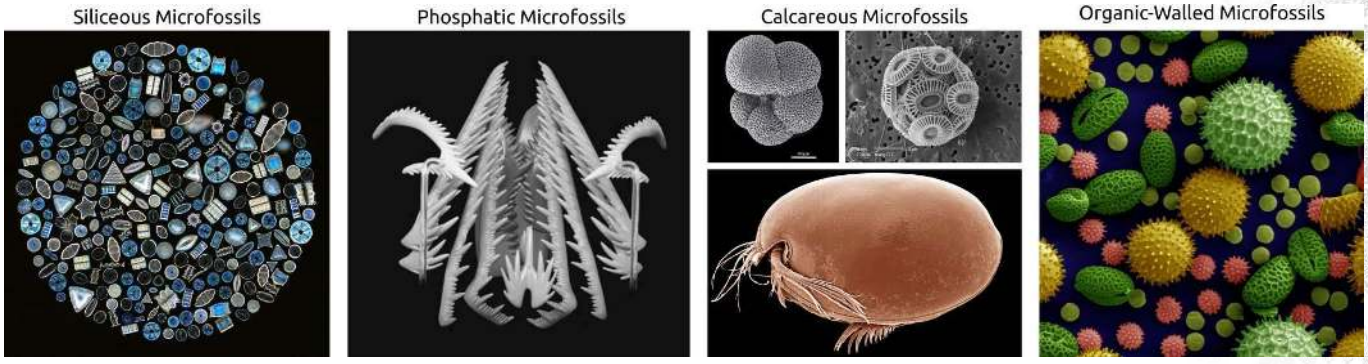
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Microfossils

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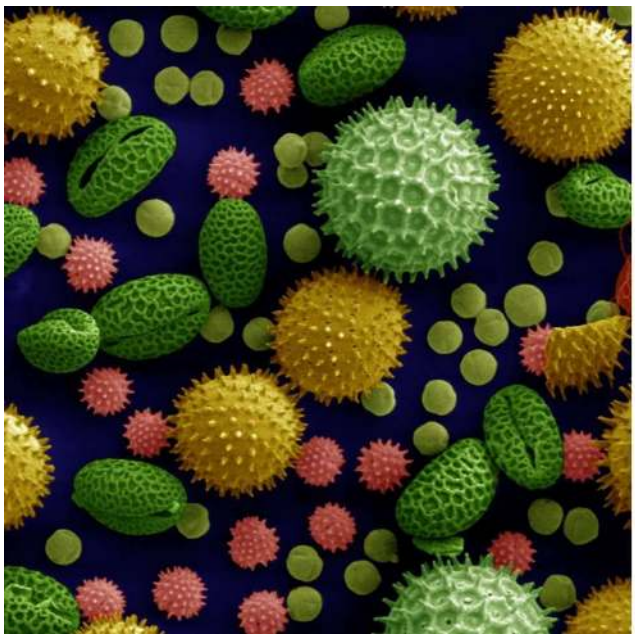
Organic-Walled Microfossils

-- Spores and Pollen (Ordovician to Recent)

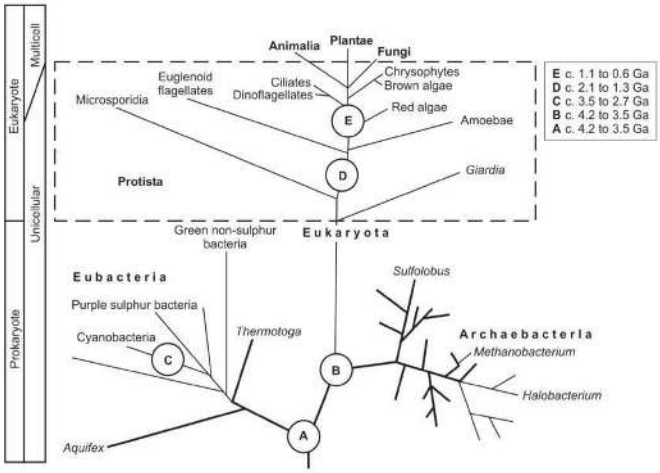
-- Dinoflagellates (Silurian to Recent)

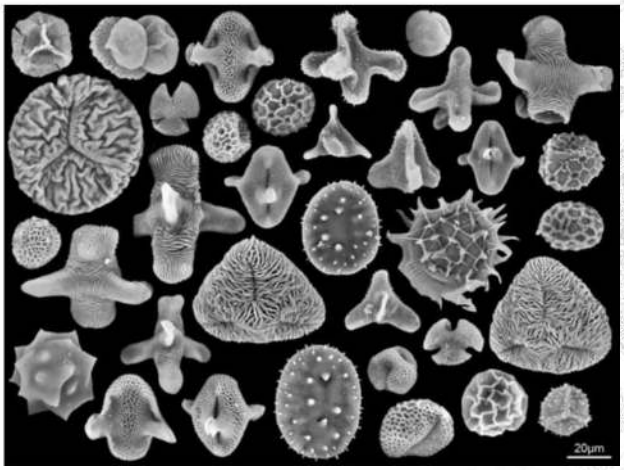
-- Acritarchs (Precambrian to Recent)
-- Tasmanitids (Cambrian to Tertiary)
-- Chitinozoa (Ordovician to Devonian)

What are spores and pollen?



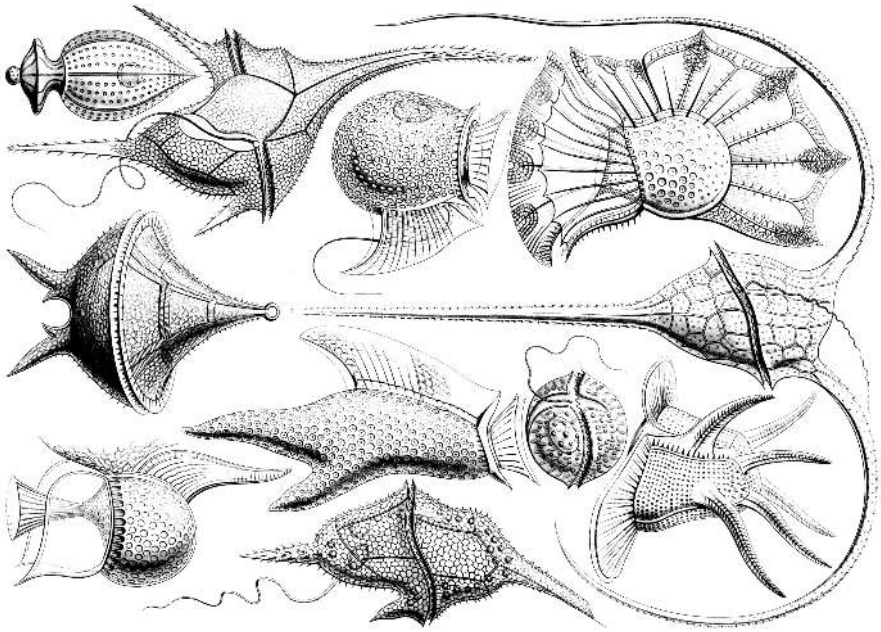






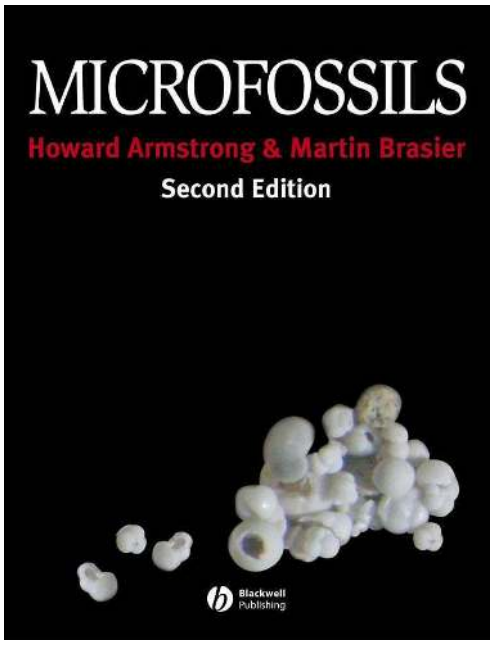
Source: Armstrong, H. and Brasier, M., 2013. Microfossils. John Wiley & Sons.

What are dinoflagellates?



Source: Haeckel, E., 1904. Kunstformen der Natur. Die einhundert Tafeln.

Want to learn more?



Armstrong, H. and Brasier, M., 2013. Microfossils. John Wiley & Sons.